

(iii) Why alloys like constantan and manganin are used for making standard resistors? 1+3+1

OR

(i) Define the capacitance of a capacitor. Obtain the expression for the capacitance of a parallel plate capacitor in vacuum in terms of plate area A and separation d between the plates.

(ii) A slab of material of dielectric constant K has the same area as the plates of a parallel plate capacitor but has a thickness $3d/4$. Find the ratio of the capacitance with dielectric inside it to its capacitance without the dielectric. 1+2+2

32.

i) State Gauss's theorem. Using this to calculate the electric field of a thin infinitely long straight line of charge with uniform charge density.

ii) A slab of material of dielectric constant K has the same area as the plates of parallel plate capacitor but has the thickness $d/3$, where d is the separation between the plates. Find the expression for the capacitance when the slab is inserted between the plates.

OR

i) A series LCR circuit is connected to an a.c source having voltage $V = V_0 \sin \omega t$. Derive the expression for the instantaneous current I and its phase relationship to the applied voltage.

ii) Obtain the condition for resonance to occur.

iii) Define power factor. State the conditions under which the power factor is maximum and minimum.

33.

(a) Derive the expression for drift velocity of free electrons in a conductor in terms of relaxation time of electrons.

(b) Prove that the current density of a metallic conductor is directly proportional to the drift speed of electrons.

OR

(a) Show, on a plot, variation of resistivity of

(i) a conductor, and

(ii) a typical semiconductor as a function of temperature.

(b) Using the expression for the resistivity in terms of number density and relaxation time,

5

5

General Instructions:
MAXIMUM MARKS: 70 MARKS

TIME ALLOWED: 3 HRS

- (1) There are 33 questions in all. All questions are compulsory.
 (2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.

(3) All the sections are compulsory.
 (4) **Section A** contains five questions, twelve MCQ and four Assertion Reasoning based of 1 mark each, each, **Section B** contains sixteen questions, two marks each, **Section C** contains seven questions of three marks long answer questions of two marks each, **Section D** contains four study-based questions of four marks each and **Section E** contains three long answer questions of five marks each.

(5) There is no overall choice. However, an internal choice has been provided in one question in Section B, one question in Section C, one question in each CBQ in Section D and all three questions in Section E. You have to attempt only one of the choices in such questions.

(6) Use of calculators is not allowed.
 (7) You may use the following values of physical constants where ever necessary

$$c = 3 \times 10^8 \text{ ms}^{-1}$$

$$h = 6.626 \times 10^{-34} \text{ Js}$$

$$e = 1.602 \times 10^{-19} \text{ C}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$$

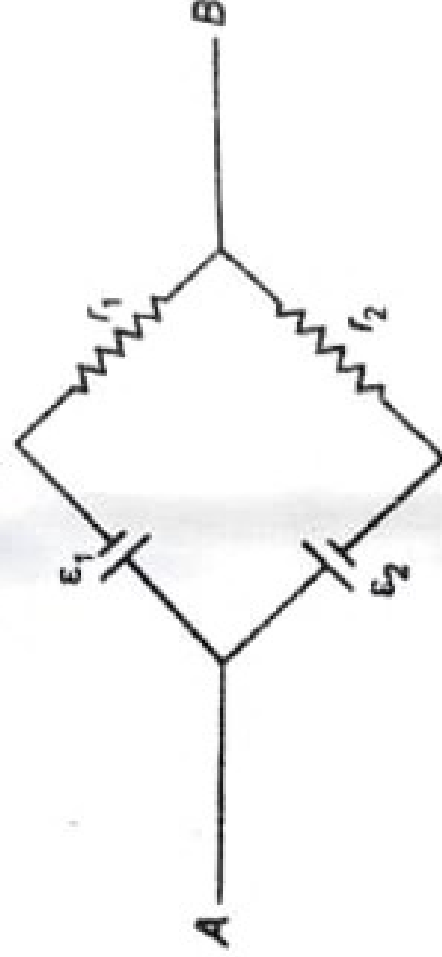
$$\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2}$$

Mass of electron $m_e = 9.1 \times 10^{-31} \text{ kg}$
 Mass of neutron $m_n = 1.675 \times 10^{-27} \text{ kg}$
 Mass of the proton $m_p = 1.673 \times 10^{-27} \text{ kg}$

Boltzmann constant $k = 1.381 \times 10^{-23} \text{ JK}^{-1}$
 Avogadro' number $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$
 Radius of earth = 6400 km

SECTION A

1. Two batteries of ϵ_1 and ϵ_2 ($\epsilon_2 > \epsilon_1$) and internal resistance r_1 and r_2 respectively are connected in parallel as shown in figure.



- (a) The equivalent emf ϵ_{eq} of the two cells is between ϵ_1 and ϵ_2 i.e. $\epsilon_1 < \epsilon_{eq} < \epsilon_2$.
 (b) The equivalent emf ϵ_{eq} is smaller than ϵ_1 .
 (c) The ϵ_{eq} is given by $\epsilon_{eq} = \epsilon_1 + \epsilon_2$ always.
 (d) ϵ_{eq} is independent of internal resistances r_1 and r_2 .

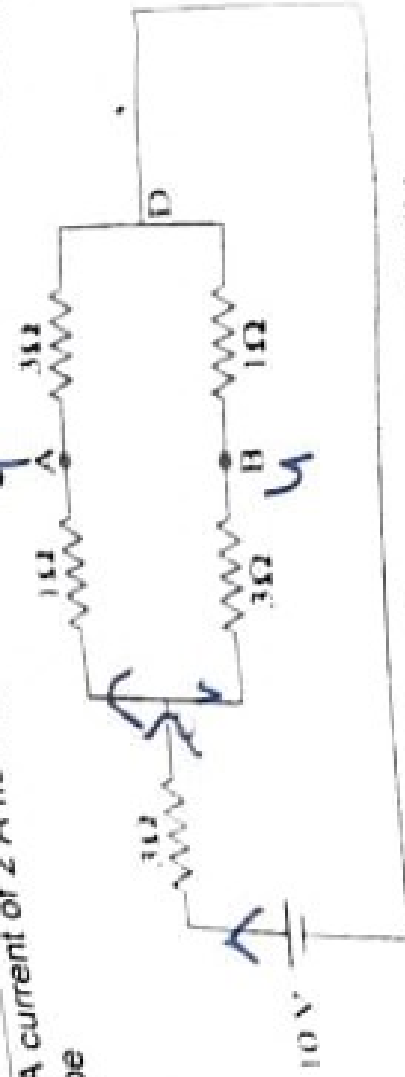
2. Biot-Savart law indicates that the moving electrons (velocity v) produce a magnetic field B such that

- (a) $B \perp v$.
 (b) $B \parallel v$.

(c) it obeys inverse cube law. (d) it is along the line joining the electron and point of observation

3. A capacitor of $50\mu\text{F}$ is connected to a supply of 220 V and angular frequency 50 rad/s. The value of rms current in the circuit is

- (a) 0.55 A
 (b) 0.50 A
 (c) 0.45 A
 (d) 0.60 A



$$V = IR$$

$$10 =$$

(d)-2V

(c)-1V

(a)+2V

(b)=1V

(d)-2V

A 50 ohm galvanometer gets full scale deflection when a current of 0.01 A pass through the coil.

When it is converted to a 10 A ammeter, the shunt resistance is

(a) 0.01 ohms (b) 0.05 ohms (c) 2000 ohms (d) 5000 ohms

If $L=100\mu F$, current changes by 1 A in 0.1 s. What is the emf produced?

(a) 1 mV (b) 100mV (c) 10mV (d) 0.1 V

The ratio of secondary to primary turns is 4:5. If power input is P , what will be the ratio of power

output to power input?

(a) 4:9 (b) 9:4 (c) 5:4 (d) 1:1

In which type of material the magnetic susceptibility does not depend on temperature?

(a) Diamagnetic (b) Paramagnetic (c) Ferromagnetic (d) Ferrite

By using only two resistance coils -singly, in series or in parallel one should be able to obtain

resistance of 3, 4, 12 and 16 ohm. The separate resistances of the coils are

(a) 3 and 4 (b) 4 and 12 (c) 12 and 16 (d) 16 and 3

n small metal drops of same size are charged to V volt each. If they coalesce to form a single

large drop, then it's potential will be

(a) V/n (b) Vn (c) $Vn^{1/3}$ (d) $Vn^{2/3}$

A neutron, a proton, an electron and an alpha particle enter a region of uniform magnetic field

with equal velocities. The magnetic field is perpendicular directed into the paper. The tracks of

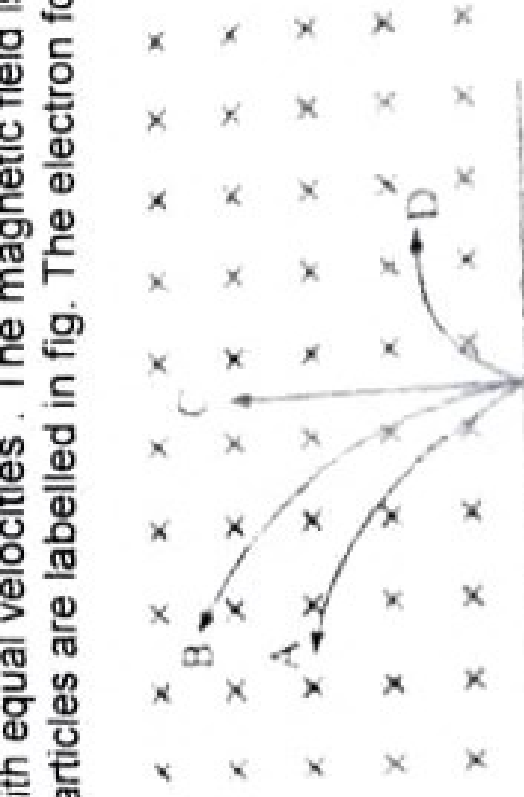
particles are labelled in fig. The electron follows track:

(a) A (b) B (c) C (d) D

A conductor of length 1 m and carrying current of 1 A is placed perpendicular to the magnetic field

of 1 Gauss. The magnetic force acting on the conductor is

(a) zero (b) 1 newton (c) 10^{-4} newton (d) 1 dyne



(a) A

(b) B

(c) C

(d) D

A conductor of length 1 m and carrying current of 1 A is placed perpendicular to the magnetic field

of 1 Gauss. The magnetic force acting on the conductor is

(a) zero (b) 1 newton (c) 10^{-4} newton (d) 1 dyne

For Questions 13 to 16, two statements are given – one labelled Assertion (A) and other

labelled Reason (R). Select the correct answer to these questions from the options as

given below.

a) If both Assertion and Reason are true and Reason is correct explanation of Assertion.

b) If both Assertion and Reason are true but Reason is not the correct explanation of

Assertion.

c) If Assertion is true but Reason is false.

d) If both Assertion and Reason are false.

Assertion (A): A charge, whether stationary or in motion produces a magnetic field around it.

Reason (R): Moving charges produce only electric field in the surrounding space.

Assertion(A) It is advantageous to transmit electric power at high voltage.

Reason (R): High voltage implies high current.

Assertion (A): A photon has no rest mass, yet it carries definite momentum.

Reason(R): Momentum of photon is due to its energy and hence its equivalent mass.

Assertion : A metallic shield in form of a hollow shell may be built to block an electric field.

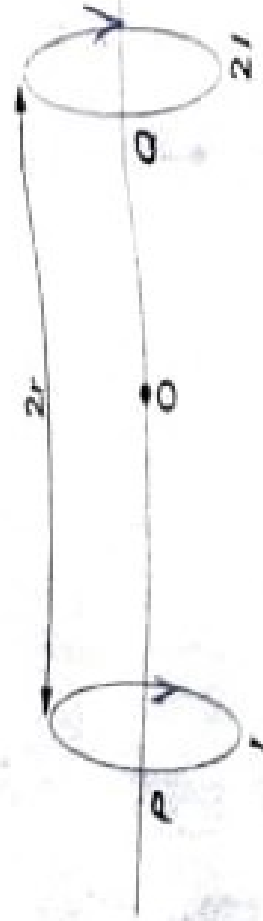
Reason : In a hollow spherical shield, the electric field inside it is zero at every point.

17. A capacitor of unknown capacitance is connected across a battery of V volts. The charge stored in it is $360\mu\text{C}$. When potential across the capacitor is reduced by 120 V , the charge stored in it becomes $120\mu\text{C}$. Calculate (i) The potential V and the unknown capacitance C . (ii) What will be the charge stored in the capacitor, if the voltage applied had increased by 120 V ? 600 2
18. A cell of emf E and internal resistance r is connected across a variable resistance R . Plot graphs showing the variation of (i) E and R (ii) terminal potential difference V with R . Predict from the second graph under which V becomes equal to E . 2
19. Two circular coils, one of radius r and the other of radius R are placed coaxially with their centres coinciding. For $R \gg r$, obtain an expression for the mutual inductance of the arrangement. $\mu_0 I r^2 / 2R$ 2
20. A proton and an alpha particle are accelerated through the same potential. Which one of the two has (i) greater value of de-Broglie wavelength associated with it, and (ii) less kinetic energy? Justify. 2

OR

Calculate the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its (i) second permitted energy level to the first level, and (ii) the highest permitted energy level to the first permitted level.

- Two identical circular loops, P and Q , each of radius r and carrying currents I and $2I$ respectively are lying in parallel planes such that they have a common axis. The direction of current in both the loops is clockwise as seen from O which is equidistant from the both loops. Find the magnitude of the net magnetic field at point O .



SECTION-C

- A proton and an alpha particle enter at right angles into a uniform magnetic field of intensity B . Calculate the ratio of the radii of their paths, when they enter the field with the (i) same momentum, (ii) same kinetic energy, (iii) same velocity. 3

- (i) Derive the expression for resistivity in terms of relaxation time. (ii) Hence plot a graph showing the variation of resistivity of a conductor and semiconductor, with the increase of temperature. 3

Two cells of emfs 1.5 V and 2 V having internal resistances $0.2\ \Omega$ and $0.3\ \Omega$ respectively are connected in parallel. Calculate the emf and the internal resistance of the equivalent cell.

- (i) Write the principle of moving coil galvanometer.

- (ii) What is the importance of radial magnetic field and how is it produced?

- (iii) Can we decrease or increase the range of an ammeter?

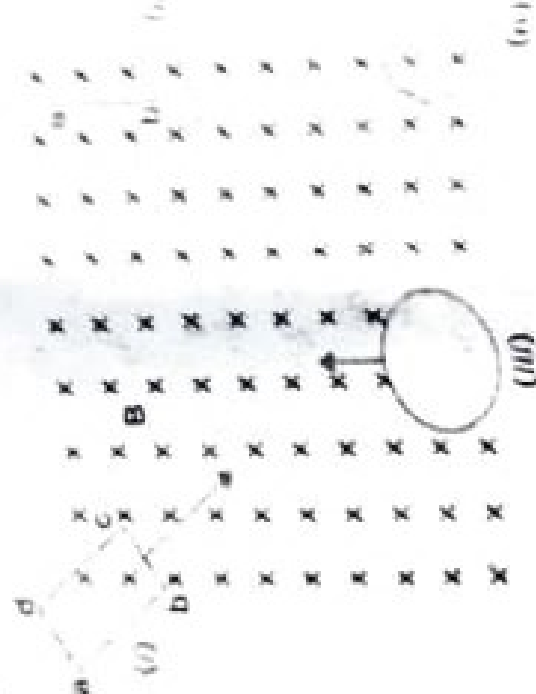
Write three points of differences between diamagnetic, paramagnetic and ferromagnetic materials.

OR

- (i) Distinguish between conduction current and displacement current.

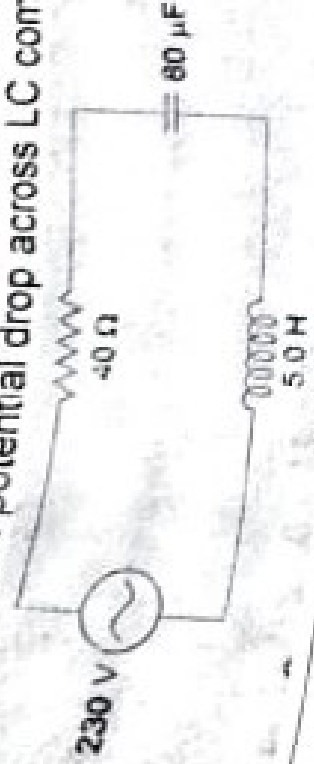
- (ii) A conductor of length 1 m falls freely under gravity from a height of 10 m so that it cuts the lines of force of the horizontal component of earth's magnetic field of $3 \times 10^{-5}\text{ T}$. Find the current induced in the conductor.

Figure shows planar loops of different shapes moving out of or into a region of magnetic field which is directed normal to the plane of loops downwards. Determine the direction of induced current in each loop using Lenz's law.



The figure shows a series LCR circuit connected to a variable frequency 230 V source.

- Determine the source frequency which drives the circuit in resonance.
- Calculate the impedance of the circuit and amplitude of current at resonance.
- Show that potential drop across LC combination is zero at resonating frequency.



29.

SECTION-D

When a current through a coil changes, the magnetic flux linked with the coil changes. As a result an emf is induced in the coil. This phenomenon is called self induction. As per Lenz's law the induced emf opposes the change of current. A solenoid made from a thick wire has negligible resistance but sufficiently large inductance. Such an element is called an ideal inductor. Inductance of a coil depends on the geometry of the coil, number of turns in the coil and permeability of the medium inside the coil.

Now answer any four from the following questions:

(i) What is the unit of self inductance of a coil?

- (a) volt s⁻¹ A⁻¹ (b) volt s⁻¹ A (c) volt² s A⁻¹ (d) volt s A⁻¹

(ii) If the number of turns per unit length of a coil of a solenoid is doubled, its self inductance will

(a) remain constant (b) be doubled (c) be halved (d) be four times.

(iii) If both the number of turns and core length of an inductor are doubled keeping other factors constant, self inductance will be

- (a) unaffected (b) double (c) halved (d) quadrupled.

(iv) In a coil of self inductance 5 henry, the rate of change of current is 2 amp per second. The emf induced in the coil is

- (a) -5 V (b) 5 V (c) -10 V (d) 10 V

(v) A coil of 100 turns carries a current of 5mA and creates magnetic flux of 10⁻³ Weber. The inductance is

- (a) 0.2 mH (b) 2 mH (c) 0.02 mH (d) none of these.

30. A wire of length 12 cm; resistance 12 Ω and of uniform area of cross-section is cut into twelve

equal parts, which are connected to form a skeleton cube. A cell of emf 2 V is connected across the two diagonally opposite corners of the cube. Using Kirchhoff's laws of junction and loop.

NOW ANSWER THE FOLLOWING QUESTIONS

(i) The effective resistance of the circuit is

- (a) 4/5 Ω (b) 5/6 Ω (c) 6/7 Ω (d) 7/12 Ω

(ii) The current drawn from the battery is

- (a) 2.5 A (b) 2.4 A (c) 2.3 A (d) 3.4 A

(iii) The maximum current flowing in an arm of network is

- (a) 0.4 A (b) 0.8 A (c) 1.2 A (d) 2.4 A

(iv) The minimum potential difference across an arm of network is

- (a) 0.4 V (b) 0.8 V (c) 1.2 V (d) 2.4 V

After cutting the wire into twelve equal parts, the resistance of each part is

- 1 Ω (b) 2 Ω (c) 3 Ω (d) 4 Ω

(v) If these twelve parts connected in parallel then the equivalent resistance will be

- (a) 0.083 Ω (b) 0.053 Ω (c) 0.090 Ω (d) 0.01 Ω